HADHRAMOUT UNIVERSITY

COLLEGE OF COMPUTERS & INFORMATION TECHNOLOGY

Midterm Exam

10/20

Academic year: 2025-2026 | **Subject:** Discrete structure

Semester: First | **Examiner:** Aisha Al-sadi

Department: IT | **Day and Date:** 6/11/2025, Thursday

Level: 1st (A) | **Time allowed:** 1 h

Name: .....

Question 1 (5 Marks):

Put True (T) or False (F) of the following sentences:

1) (F) "Write your homework" is a proposition.
2) (T) The proposition is either true or false and not both.
3) (F) If $0 > 1$ then $2 > 1$ is true.
4) (T) The inverse of the function f is defined if f is bijection.
5) (T) The set A is a subset of the set B If all element of A is also element of B

Question 2 (15 Marks):

1) If $A = \{3,4\}, B = \{1,2,3\}, C = \{0,2,4,\ldots\}$, where U is all positive integers, find

a) Write the set (C) by special property method. $C = \{x \mid x$ is an even integers $\}$

b) $\bar{C}$ = is all integers in Z / C set = Z - C or even integer.

c) $A \times B = \{(3,1),(3,2),(3,3),(4,1),(4,2),(4,3)\}$

d) $A = B$? = $\{4\}$

2) **Prove the following:**

a) If m and n are integers and mn is even, then m is even or n is even. (indirect)

b) Determine whether the function is **onto**, where $(f: \mathbb{R}^* \to \mathbb{R})$ s.t $f(x) = 1/x^3$.

a) let m is even integer. $= m^2 = (2k)^2$ $= m^2 = 2(n)$ $\therefore n = k^2$

$= m^2 = 4k^2$ let $k^2 = n$

$\therefore$ is even integer

b) let $(F: \mathbb{R}^* \to \mathbb{R})$ let $F(x) = \frac{1}{x^3} = F(1) = \frac{1}{1^3} = 1$

is ~~not onto~~ is onto